8D report

Customer plant:	AVO Tianjing	**Supplier:**	AVO Carbon Germany	**Date of claim:**	13.10.2025
Customer contact:	Weijiang Peng	**AVO contact:**	G. Luschtinetz		
Customer reference:	13.10.2025	**AVO reference:**	251009		

QUICK RESPONSE (D1-D3)

This document must be issued within 24 hours after claim notification

D 1. Description of the problem

1.a What is the problem? Use 5W+2H to describe what is the current situation.

Problem description	Copper wire fell off and serious appearance defects during the grinding process	What happened	Copper wire fell off and serious appearance defects during the grinding process
		Why is it a problem	Parts can`t be assembled. Parts out of spec.
Incident type	Customer claim		
Part number AVO	5000621	When did it happen	14.10.2025
Part number Customer	5000621	Who detected it	Inspector
Batch number	various	Where was it detected	Production
Recurrence	Non-recurrent	How was it detected	Visually
Defect S/R	No	How many bad parts	79
Standard process used	Yes		
When was the part produced?	Pressing machine => AVO Frankfurt Grinding machine => AVO Tianjing		

nOK part picture/ sketch

OK part picture/ sketch

1.b Problem solving team

Name	Position	Location	E-mail
G. Luschtinetz	Quality Management	AVO Germany	Gunther.luschtinetz@avocarbon.com
J.Gehl	Quality Management	AVO Germany	Joachim.Gehl@avocarbon.com
M. Yorulmaz	Production	AVO Germany	
G. Balenzano	Production	AVO Germany	
K. Amri	Logistic	AVO Germany	

1.c Tools to use

Yes	Ishikawa	Yes	5why's		LLC		Audit		Other

1.d First analyse of problem / potential root causes

Potential root cause(s): One of the grub screw has loose and the punch lost his fixing

If you use our expressnumber, please use only economic express.

Samples of nok parts required in plant for analyse: Yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts?		Comments (compulsory, if yes)
Other family product	No	
Other line	No	
Other tools/work station	No	
Other	No	

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	N° of involved parts	N° parts sorted	N° nok parts	Ppm	Date	Responsible
Parts at customer	702.075	701.996	79	---	22.10.2025	Weijiang Peng
Parts in transport	---	---	---	---	---	---
Parts at AVO Germany stock		---	---	---		
Parts at AVO production	---	---	---	---	---	---

First contained shipment information

Immediate actions to prevent production and delivery of nok parts ?

Containment actions	Responsible	Date planned	Date closed
Register the complaint + Quality alert	G. Luschtinetz	14.10.2025	14.10.2025
Check the production-, quality- and maintenance data	G. Luschtinetz	14.10.2025	14.10.2025
Information of all department leader during shop floor meeting QRQC	D. Beil	14.10.2025	14.10.2025
Sorting parts at customer (7778,2 €)	W. Peng	14.10.2025	22.10.2025
Permanent sorting (100 %)	D. Beil	27.08.2025	27.08.2025

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PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

The punch has different single parts, the mounting and five punch inserts. Every punch insert is fixed with a single grub screw. One of the grub screw has loose and the punch lost his fixing. As the result, the punch insert has been pushed upwards and the wire has not been pressed in deep enough. Based on the fewer inseration deepth of the wire, the wire come loose.
Measures were not implemented for all tools.

How was the defect reproduced ?

Fix a punch insert not properly.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

The defect cannot be recognized by the measurements during the hourly check, as it occurs sporadically.

D 6. Corrective actions and non detection

Corrective actions	Responsible	Date planned	Date closed
Re-design the punches without punch inserts	U. Wallocha	14.10.2025	31.10.2025
All drawings converted to upper punches without inserts	U. Wallocha	14.10.2025	03.11.2025
Scrap of punches with old design	U. Wallocha	17.12.2025	
Order punches with new design	U. Wallocha	16.12.2025	

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Responsible	Date planned	Date closed
During the next workplace audit - tool shop especially: are only the new punches stored	Team	17.12.2025	

D 8. Lessons learned

8.1 Do following documents require an update ?

	Responsible	Date planned	Date closed
P-FMEA	Team	18.12.2025	
Control Plan			
Procedure			
Instructions			
Other			
Other			

Generally, if one of the tools listed above is used. This is carefully checked and updated.

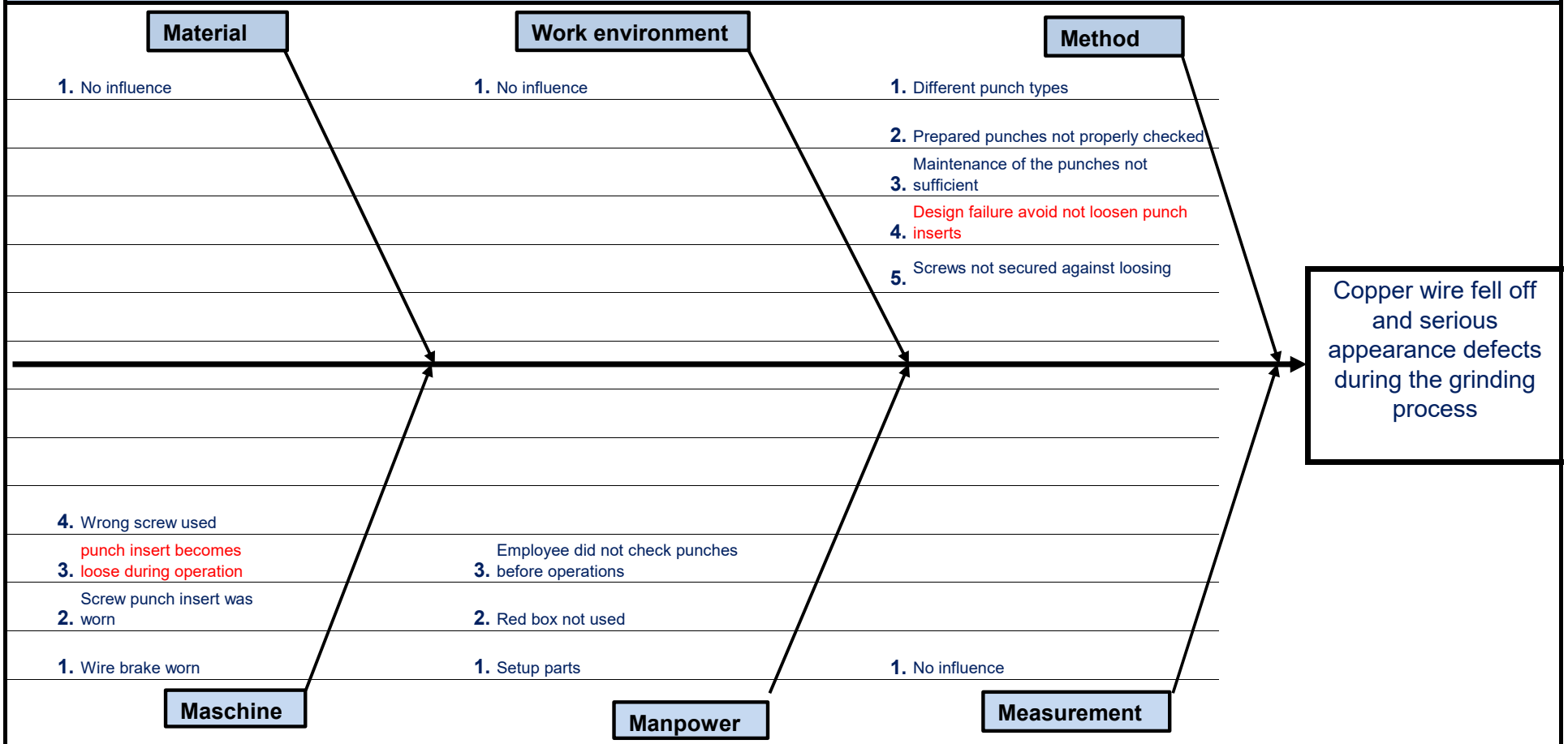
8.2 Generalization of countermeasures

8.3 Internal audit (if required)

Date | Results: ok

Comments

Ishikawa


Material
1. No influence
Work environment
1. No influence
Method
1. Different punch types
2. Prepared punches not properly checked
3. Maintenance of the punches not sufficient
4. Design failure avoid not loosen punch inserts
5. Screws not secured against loosing
Copper wire fell off and serious appearance defects during the grinding process
4. Wrong screw used
3. punch insert becomes loose during operation
2. Screw punch insert was worn
1. Wire brake worn
Maschine
3. Employee did not check punches before operations
2. Red box not used
1. Setup parts
Manpower
1. No influence
Measurement

5 Why's

Problem description : Copper wire fell off and serious appearance defects during the grinding process

OCCURENCE

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY?	5th ANSWER
	1	Why is the wire loosen?	Because the wire depth is not according to the specification.	Why is the wire not according to the specification?	Because the wire was not fixed during pressing.	Why was the wire not fixed during pressing?	Because a punch insert was loosen.	Why was the punch insert loosen?	Because a screw was not tighten enough.	Why was the screw not tighten enough?	Because the screw had loosen during operation.
	N°	6th WHY?	6th ANSWER								
	1	Why had the screw become loosen during operation?	Because the screw was not secured against loosing.								

NON-DETECTION

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY ?	5th ANSWER
	1	Why was the defect not detected?	Because the visual control did not find the failure.	Why did not find the visual control the failure?	Because ist just visual check.	Why is just visual checked?					

D 6. Corrective actions and non detection			
Corrective actions	Responsible	Date planned	Date closed
Re-design the punches without punch inserts	U. Walocha	14.10.2025	
Measures finally implemented for all tools.	U. Walocha	14.10.2025	

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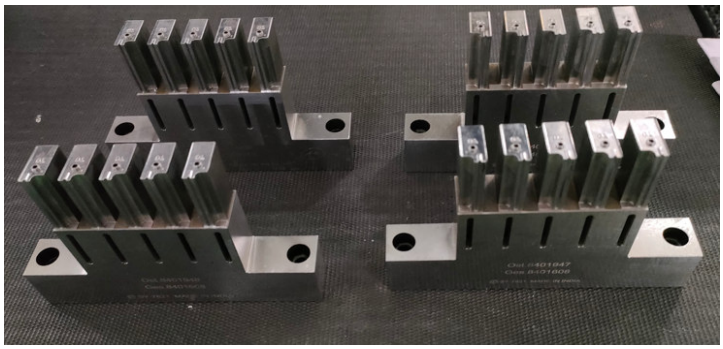

Two upper punches without inserts still remain open

2 x 8401947 Oberstempelkamm geliefert 06.12.2024
2 x 8401948 Oberstempelkamm geliefert 06.12.2024
2 x 8401947 Oberstempelkamm geliefert 14.04.2025
2 x 8401948 Oberstempelkamm geliefert 14.04.2024
3 x 8401707 Oberstempelkamm geliefert 22.09.2025
3 x 8401709 Oberstempelkamm geliefert 22.09.2025
2 x 8401624 Oberstempelkamm noch nicht freigegeben.117481/OR
2 x 8401609 Oberstempelkamm noch nicht freigegeben.117481/OR